

The single most popular and widely implemented firmware TPM (fTPM) module used among IoT and embedded devices is the **Microsoft TPM 2.0 Reference Implementation** (commonly referred to as ms-tpm-20-ref). [1, 2, 3]

While it originated from Microsoft, it serves as the official reference implementation for the **Trusted Computing Group (TCG)** TPM 2.0 Library Specification. It is entirely open-source, written in C, and acts as the foundational backbone for virtually all modern, non-discrete embedded trusted applications. [1, 4, 5, 6, 7]

Why ms-tpm-20-ref Dominates the Embedded Landscape

Embedded and IoT platforms utilize this reference code by adapting it to run inside a **Trusted Execution Environment (TEE)**, leveraging secure hardware features like **ARM TrustZone**. This architecture provides a massive cost savings for high-volume IoT products because it implements robust security without requiring a physical, discrete dTPM microchip. [8, 9, 10, 11, 12]

The module is integrated and packaged across the embedded ecosystem through several notable deployments: [13, 14]

- **OP-TEE fTPM (The De Facto Linux Standard):** In the embedded Linux and Yocto Project ecosystem, the reference code is compiled as a Trusted Application (TA) using the OP-TEE (Open Portable TEE) framework. It is widely used via the [meta-arm Yocto layer](#) to bring native fTPM capabilities to custom Linux distributions running on IoT gateways and industrial equipment. [4, 15, 16, 17, 18]
- **NVIDIA Jetson Platforms:** Heavy-duty edge AI and embedded platforms like the **NVIDIA Jetson AGX Orin, Orin NX, and Orin Nano** explicitly use this TCG reference implementation integrated into OP-TEE to handle secure boot, credential protection, and remote attestation. [18, 19, 20, 21, 22]
- **Commercial SDK Wrappers:** Leading embedded security vendors wrap this reference code into production-ready commercial frameworks. For example, [SecEdge's SEC-TPM](#) pairs it with automated provisioning tools for STMicroelectronics platforms, and [wolfSSL's wolfTPM](#) provides optimized bare-metal API wrappers around it for highly constrained microcontrollers. [5, 23, 24]

Silicon-Vendor Specific Alternatives

If you look at broad consumer and commercial silicon spaces that cross into "heavy" IoT (like industrial PCs or smart kiosks), vendors also maintain highly optimized, proprietary versions of fTPM directly baked into their firmware: [11, 25]

- **Intel PTT (Platform Trust Technology):** Intel's proprietary fTPM that runs on the Management Engine (ME) across x86-based IoT and edge computers.
- **AMD fTPM:** AMD's equivalent architecture, which operates within the Platform Security Processor (PSP). [25, 26, 27, 28, 29]

If you are currently designing or deploying an ARM-based IoT device and looking for the easiest

way to proceed, **building on top of the OP-TEE fTPM (Microsoft reference-backed) stack** is the industry standard for vendor-neutral development. [4, 30]

To help tailor a solution, let me know: What **system-on-chip (SoC)** or hardware platform are you developing on? Also, what **operating system** (e.g., Embedded Linux, Zephyr RTOS, Windows IoT) will the device run?

- [1] <https://github.com>
- [2] <https://docs.nvidia.com>
- [3] <https://docs.nvidia.com>
- [4] <https://layers.openembedded.org>
- [5] <https://www.wolfssl.com>
- [6] <https://en.wikipedia.org>
- [7] <https://www.microsoft.com>
- [8] <https://trustedcomputinggroup.org>
- [9] <https://www.usenix.org>
- [10] <https://patents.google.com>
- [11] <https://www.indurock.com>
- [12] <https://www.assured-systems.com>
- [13] <https://www.linkedin.com>
- [14] <https://community.intel.com>
- [15] <https://archive.fosdem.org>
- [16] <https://github.com>
- [17] <https://crosscon.eu>
- [18] <https://docs.nvidia.com>
- [19] <https://sergioprado.blog>
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- [23] <https://www.st.com>
- [24] <https://www.secedge.com>
- [25] <https://premioinc.com>
- [26] <https://www.bvm.co.uk>
- [27] <https://www.computer.org>
- [28] <https://github.com>
- [29] <https://news.ycombinator.com>
- [30] <https://crosscon.eu>